

ASAD HUSSAIN

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PROFESSIONAL SUMMARY

Final-year Electrical Engineering student at NUST working across applied machine learning and embedded/RF systems engineering. Combines backend/AI engineering (FlyRank AI), applied ML (CodeAlpha), and deep learning/LLM/RAG training (AtomCamp) with hands-on RF hardware and embedded firmware experience, including DroneGuard, a two-board RF detection and jamming system, and IoT telemetry work from prior internships. Comfortable moving between training models and building the circuits and firmware that feed them real-world data.

TECHNICAL SKILLS

Programming: Python · C++/Embedded C · JavaScript · SQL · MATLAB/Simulink · Flask

ML / AI: Scikit-learn · TensorFlow · PyTorch · LangChain · RAG · LLM Fine-tuning · FastAPI · Docker

Hardware & RF: SDR · STM32 · Arduino · Raspberry Pi · PCB Design (Proteus) · Antenna Design

Tools: Git · REST APIs · SQL Server · Google Cloud · Weights & Biases · AutoCAD · Altium Designer

WORK EXPERIENCE

Technical Intern | National Electric Power Regulatory Authority (NEPRA), *Islamabad*

Jul – Aug 2026

- Studied NEPRA's end-to-end regulatory flow (license → tariff → sale) across Generation, Transmission, Distribution, and Supply licenses, including the three generation tariff mechanisms.
- Analyzed NGC's transmission network data, 233 grid-station transformers (116 above 80% rated capacity), and NEPRA's State of the Industry Report, covering 111,466 GWh of national consumption.

Backend AI Engineering Intern | FlyRank AI, *Remote, Chicago, USA*

Jun – Aug 2026

- Completed a backend/AI engineering practicum: built an authentication-protected CRUD API with a connected database, containerized the stack with Docker, and automated a data-scraping/workflow pipeline.
- Applied prompt-engineering and LLM-tooling fundamentals to search-intelligence tasks (baseline scoring and top-10 review pipeline) across 21 practical assignments and 19 workshops on AI tooling and backend development.

Machine Learning Intern | CodeAlpha, *Remote*

May – Jun 2026

- Built a credit-scoring classification model (Logistic Regression, Decision Trees, Random Forest) with feature engineering from financial history data, evaluated using Precision, Recall, F1-score, and ROC-AUC.
- Built a CNN-based handwritten character recognition model (MNIST/EMNIST) and a disease-prediction classifier on structured medical data (symptoms, age, lab results) using SVM, Logistic Regression, and Random Forest.

Engineering Intern, Full-Stack Systems | Research & Indigenous Development Centre (RDC), HIT

Jun – Aug 2025

- Integrated core hardware (LilyGO T-SIM7670E, T3-S3 GPS, RDM6300 RFID) with C/C++ state machines for GPS capture, RFID authentication, and MQTT telemetry over cellular GPRS.
- Built an asynchronous Python Flask backend with multithreading to process analytics workloads without blocking incoming MQTT streams.

PROJECTS

DroneGuard — Anti-Drone Surveillance System

2025 – 2026

SDR · GNU Radio · Raspberry Pi 4 · STM32H743 · MATLAB · Python

- Designed a two-board anti-drone system: a four-band RF detection front-end (433 MHz–2.4 GHz) and a jamming board built around an ADF4351 PLL synthesiser.
- Built a real-time signal-classification pipeline (RTL-SDR + GNU Radio) identifying FHSS, OFDM, and FSK drone protocols from live IQ samples, plus a MATLAB front-end for live spectrum and telemetry.

Academic Projects

- Hands-on projects across core EE disciplines: Butterworth Active LPF (op-amp) · Multi-Stage Audio Amplifier · EEG Cleaning MATLAB GUI · PID DC Motor Controller (Arduino) · 5-Band Graphic Equaliser (Simulink) · 4-Bit ALU · FM Receiver (TDA7000) · Multi-Range Voltmeter · OOP Shopping Cart (C++) · Water Level Indicator (Proteus) · Current Transformer

TRAINING

- AtomCamp AI Bootcamp — Deep Learning, NLP/LLMs, n8n, Generative AI & RAG (LangChain agent development), and MLOps/deployment (Docker, FastAPI, CI/CD, Google Cloud).

EDUCATION

B.E. Electrical Engineering | National University of Sciences and Technology (NUST)

2023 – 2027

- Coursework: Analog Electronics, DSP, Control Systems, Communication Systems, Digital Logic Design, Embedded Systems, Signals & Systems, Instrumentation, OOP, Microprocessor Systems

CERTIFICATIONS, ACTIVITIES & ACHIEVEMENTS

- MATLAB Fundamentals (MathWorks) · PCB Design (Proteus & SPICE) · Technical Lead, NUST Robotics Society · Highest Achiever, Matriculation (SSC) · 1st Place Robotics, Voltfest · NMUN 2024 Delegate
- Portfolio:— [Asad Hussain | Electrical Embedded Engineer](#) full-stack showcase of embedded, RF/SDR, and ML/AI projects.